

Homework 4

STAT 300 (Fall 2024)

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Clearly label your answers to each question and each sub-question. Your answers MUST be uploaded to Canvas as a <HWx_Yourfirstname.nb.html> file by the deadline.

- (1). For each of the below, state if it is true or false. If it is false explain the reason.
 - (a) If a dataset A has a larger correlation between Y and X than dataset B, then the slope between Y and X for dataset A will be larger than the slope between Y and X for dataset B.
 - (b) The model degrees of freedom is always 1 for simple linear regression.
 - (c) The magnitude of the multiplier ($t_{\frac{\alpha}{2}, (n-2)}$) used to compute the confidence interval for the slope of a regression line decreases as the sample size increases.
 - (d) The unexplained variation (SSE) is always smaller than the explained variation (SSModel)
 - (e) If the correlation between X1 and Y is greater (in magnitude) than the correlation between X2 and Y, then the coefficient of determination of the regression model of Y on X1 is greater than the coefficient of determination of the regression model of Y on X2.
- (2). A regression equation was fit to a dataset for which the correlation r between X and Y was 0.6. Which of the following must be true?
 - (a) The slope of the regression line is 0.6.
 - (b) The regression model explains 60% of the variability in Y.
 - (c) The regression model explains 36% of the variability in Y.
 - (d) At least half of the residuals are smaller than 0.6 in the absolute value.
- (3). The data *TxtPrc.csv* file has information for a random sample of 30 textbooks from Cal Poly campus bookstore. Suppose a student is interested in predicting the price based on the number of pages in the book.
 - (a) Make a scatter plot to display the relationship. Comment on your observations.
 - (b) Obtain the least squares regression model for predicting the price from the number of pages.
 - (c) Write the equation of the least squares regression line.
 - (d) Calculate the standardized residuals and store the residuals. Call it *TxtStdResid*.
 - (e) Make a plot of standardized residuals Vs. fitted values and the normal QQ plot of the standardized residuals.
 - (f) Comment on the validity of the regression assumptions.
 - (g) The student is interested in seeing if the number of pages is a significant linear predictor of the textbook price.
 - Write down the null and alternative hypotheses to answer this question.
 - You will find the numerical value of the test statistic in the R output. Show how this value is calculated from the values of $\hat{\beta}_1$ and $\hat{SE}(\hat{\beta}_1)$.

- What is the p value for this test?
 - What is your conclusion at 5% significance level.
- (h) Find a 95% confidence interval for the expected price change for every additional one page a book has. Show your mathematical work.
- (i) Repeat the question (h) using R.
- (j) Find a 95% confidence interval for the expected price change for every additional 100 pages a book has. Show your mathematical work.
- (4). Suburban towns often spend a large fraction of their municipal budgets on public safety (police, fire, and ambulance) services. A taxpayers' group felt that very small towns were likely to spend large amounts per person because they have such small financial bases. The group obtained data on the per capita expenditure for public safety of 29 suburban towns in a metropolitan area, as well as the population of each town in units of 10,000 people. The data are given in *expenditure.csv*.
- (a) Identify the explanatory and response variable in the problem.
 - (b) If the taxpayer's group is correct, what sign should the slope of the regression model have?
 - (c) Obtain a scatter plot for the problem, and comment on your observations.
 - (d) Fit a regression line for the problem. Does the slope in the output confirms the opinion of the group? Explain.
 - (e) There seems to be an unusual observation (town) in upper right corner of the scatter plot. Plot the Cook's distances for this data. Does this observation appears to be an influential point. Explain.
 - (f) Obtain the least squares regression line after removing the unusual observation. Write down the equation of the fitted line.
 - (g) Test if there is supporting evidence towards taxpayers' group belief (use the data without the unusual point)
- (5). Do a quick Google search on "extrapolation in regression". Write what is extrapolation in **your words** and explain the consequences.